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An Analysis of Yield Curve Modeling

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1 Executive Summary

This project investigates the use of time series models for forecasting the U.S. Treasury yield curve. Nelson and Siegel [Nelson and Siegel, 1987] introduced a parametric approach that effectively captures the level, slope, and curvature of interest rates. Specifically, we apply the Dynamic Nelson-Siegel (DNS) model within a state-space framework to estimate and forecast yield curve movements over time. The Kalman Filter (KF) is used as the primary tool to estimate the unobserved latent factors—level, slope, and curvature—that govern the term structure. To improve the model’s adaptability to changing market conditions, we extend the standard KF by incorporating a GARCH(1,1) process into the state noise structure, resulting in a volatility-adjusted version: KF-GARCH.

We compare the predictive accuracy and uncertainty quantification of these two models using real yield data provided by the St. Louis Federal Reserve. Through log-likelihood diagnostics and visualizations, we demonstrate that the KF-GARCH model better captures volatility clustering and offers more reliable confidence intervals, particularly for longer maturities. Our analysis highlights the importance of incorporating time-varying volatility when modeling financial time series, and lays the groundwork for more advanced forecasting techniques in dynamic risk environments.

2 Introduction and Motivation

In fixed-income market, bonds play an important role in building balanced investment portfolios, especially for long-term investors. They provide steady income and stability, helping to offset volatility of the stock market while reducing overall portfolio risk. Therefore, in our study, we will focus on building the model to capture the term structure of yield curve to help us better forecast yield to maturity and covariance matrix of yield to maturities.

A key limitation of the static Nelson-Siegel model lies in its inability to capture the dynamic evolution of the yield curve over time. While the model effectively describes the cross-sectional shape of the curve at a single point using level, slope, and curvature factors, it treats these parameters as fixed rather than time-varying.

In this project, we begin by introducing the Nelson-Siegel model and its dynamic extension, the Dynamic Nelson-Siegel (DNS) model. Following that, we present how the standard Kalman Filter (KF) can be applied to the DNS framework for yield curve modeling. To improve the model’s ability to account for changing market uncertainty, we then extend the standard Kalman Filter by integrating it with a GARCH model, resulting in the KF-GARCH approach, which allows for time-varying volatility in the state equations.

After introducing these models and the associated methods, we explain the general structure of the Kalman Filter, including the three main steps: state prediction, observation prediction, and state update. We then detail the parameter estimation procedures for both the standard Kalman Filter and the KF-GARCH model.

Finally, we present a case study using real yield curve data to compare the performance of the two models. The results highlight the advantages of modeling volatility dynamics, demonstrating that the KF-GARCH model provides improved predictive accuracy and a more realistic quantification of uncertainty.

3 Problem Statement

The work of Bliss and Fama [Fama and Bliss, 1987] introduced the smoothed bootstrap method for estimating the term structure of interest rates, leveraging segmented linear interpolation between market quotes to construct a smooth zero-coupon yield curve. Due to the limitation of low liquidity and sparsity, McCulloch [McCulloch, 1971] first introduced quadratic and cubic splines for fitting the yield curve. [Shea, 1984] and Steeley [Steeley, 1991] employed B-splines for yield curve estimation, which enhanced smoothness and flexibility while minimizing the risk of overfitting.

However, since splines rely on flexible, piecewise functions to fit yield curves, Nelson and Siegel [Nelson and Siegel, 1987] introduced a structured approach employing a parametric model that significantly advanced the field of yield curve estimation.

For the estimation of its parameters, Diebold and Li [Diebold and Li, 2006] employed OLS by fixing the shape parameter λ to transform the nonlinear problem into a linear regression task. However, since OLS is sensitive to heteroskedasticity and serial correlation in financial data, resulting in biased and inefficient estimates [Christensen et al., 2011], researchers started using Generalized Least Squares (GLS) for parameter estimation.

To incorporate time dynamics and improve predictive power, We could also placed the Nelson-Siegel model into a state-space form, allowing estimation via the Kalman Filter. This framework treats the level, slope, and curvature components as latent state variables that evolve stochastically over time. The Kalman Filter offers recursive estimation and optimal forecasting in the presence of noise and unobserved components, making it especially well-suited for yield curve modeling.

Building upon this, recent research has explored extending the Kalman Filter by incorporating time-varying volatility in the state equations. One possible enhancement involves modeling the state noise covariance matrix using a GARCH

process, allowing the model to account for volatility clustering and heteroskedasticity in interest rate dynamics. This hybrid model, often referred to as KF-GARCH, provides more realistic confidence intervals and improved prediction accuracy in turbulent market conditions

4 Model and Methods

4.1 Dynamic Nelson Siegel Model

Nelson and Siegel [Nelson and Siegel, 1987] introduced a parametric approach that significantly advanced yield curve estimation. However, static parameters may fail to capture the yield curve's evolution under time-varying economic conditions. To address this, the Dynamic Nelson-Siegel model introduces time-varying factors, enhancing its ability to track yield curve dynamics.

$$\mathbf{Y}_t(\boldsymbol{\tau}) = \mathbf{C}(\lambda; \boldsymbol{\tau})\mathbf{X}_t + \boldsymbol{\epsilon}_t, \quad (1)$$

where:

- $\mathbf{Y}_t \in \mathbb{R}^{n \times 1}$: $\mathbf{Y}_t = [Y_t(\tau_1), \dots, Y_t(\tau_N)]^T$ is the observed yield vector at different maturities $\boldsymbol{\tau}$.
- $\mathbf{C}(\lambda, \boldsymbol{\tau}) \in \mathbb{R}^{n \times 3}$: The Nelson-Siegel design matrix

$$\mathbf{C}(\lambda, \boldsymbol{\tau}) = \begin{bmatrix} 1 & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} - e^{-\lambda\tau_1} \\ 1 & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} - e^{-\lambda\tau_2} \\ \vdots & \vdots & \vdots \\ 1 & \frac{1-e^{-\lambda\tau_n}}{\lambda\tau_n} & \frac{1-e^{-\lambda\tau_n}}{\lambda\tau_n} - e^{-\lambda\tau_n} \end{bmatrix}$$

- $\boldsymbol{\beta}_t \in \mathbb{R}^{3 \times 1}$: $\mathbf{X}_t = [L_t, S_t, C_t]^T$ is the state vector at time t , representing the level, slope, and curvature factors.
- $\boldsymbol{\epsilon}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{Q})$: The process noise, where $\mathbf{R} \in \mathbb{R}^{n \times n}$ is the covariance matrix of the state noise.
- $\lambda \in \mathbb{R}$ Decay parameter controlling the yield curve's shape, which is a tuning parameter.

Extending this model, the individual yield component in vector form for a given maturity τ_i is:

$$Y_t(\tau_i) = L_t + S_t \frac{1 - e^{-\lambda\tau_i}}{\lambda\tau_i} + C_t \left(\frac{1 - e^{-\lambda\tau_i}}{\lambda\tau_i} - e^{-\lambda\tau_i} \right) + \varepsilon_t(\tau_i),$$

4.2 State Space Model

Kalman Filter widely used in financial modeling due to its ability to estimate latent state variables in dynamic systems. Its application in term structure modeling has been particularly influential, with Diebold and Li [Diebold and Li, 2006] extending the classic Nelson-Siegel yield curve model into a Dynamic Nelson-Siegel framework. Unlike the static NS model, the DNS approach treats the level, slope, and curvature factors as time-varying state variables, which evolve according to a latent process and are estimated using the Kalman filter. This technique improve the term structure modeling compare to the case where the parameter is constant over tim

$$\mathbf{X}_t = \mathbf{A} \mathbf{X}_{t-1} + \mathbf{B} \mathbf{W}_t \quad \mathbf{W}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{Q}), \quad (2)$$

$$\mathbf{Y}_t = \mathbf{C}(\lambda; \boldsymbol{\tau}) \mathbf{X}_t + \mathbf{D} \mathbf{Z}_t, \quad \mathbf{Z}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{R}) \quad (3)$$

- $\mathbf{X}_t \in \mathbb{R}^{3 \times 1}$: $\mathbf{X}_t = [L_t, S_t, C_t]^T$ is the state vector at time t , representing the level, slope, and curvature factors.
- $\mathbf{A} \in \mathbb{R}^{3 \times 3}$: The state transition matrix.
- $\mathbf{B} \in \mathbb{R}^{3 \times p}$: State noise transformation matrix, dimension correction for $\mathbf{W}_t$. (It is identity if the dimension of $\mathbf{W}_t$ is the same as $\mathbf{X}_t$).
- $\mathbf{W}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{Q})$: The process noise, where $\mathbf{Q} \in \mathbb{R}^{3 \times 3}$
- $\mathbf{Y}_t \in \mathbb{R}^n$: $\mathbf{Y}_t = [Y_t(\tau_1), \dots, Y_t(\tau_n)]^T$ is the observed yield vector at different maturities τ .
- $\mathbf{C} \in \mathbb{R}^{n \times m}$: $\mathbf{C}(\lambda, \boldsymbol{\tau})$ is the $n \times 3$ design matrix of Nelson Siegel basis.
- $\mathbf{D} \in \mathbb{R}^{n \times r}$: Observation noise transformation matrix, dimension correction for $\mathbf{Z}_t$. (It is identity if the dimension of $\mathbf{Z}_t$ is the same as $\mathbf{Y}_t$)
- $\mathbf{Z}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{R})$: The measurement noise, where $\mathbf{R} \in \mathbb{R}^{r \times r}$

4.3 State Space Model with Volatility

In traditional state-space models, the covariance matrix of the state noise is often assumed to be constant over time. However, in financial applications like yield curve modeling, this assumption might be violated. Economic and market conditions might also cause the time-varying uncertainty in the evolution of latent factors (like level, slope, and curvature) are not constant. Hence, rather than assuming the covariance matrix of the state noise to be constant, we model its time-varying behavior. Specifically, the volatility of the latent state noise is

assumed to follow a GARCH process.

$$\mathbf{X}_t = \mathbf{A} \mathbf{X}_{t-1} + \mathbf{B} \mathbf{W}_t, \quad \mathbf{W}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{Q}_t), \quad (4)$$

$$\mathbf{Y}_t = \mathbf{C} \mathbf{X}_t + \mathbf{D} \mathbf{Z}_t, \quad \mathbf{Z}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{R}), \quad (5)$$

$$\mathbf{Q}_t = \boldsymbol{\omega}_0 + \sum_{j=1}^q \boldsymbol{\alpha}_j \mathbf{W}_{t-j} \mathbf{W}_{t-j}^\top + \sum_{i=1}^p \beta_i \mathbf{Q}_{t-i} \quad (6)$$

- $\mathbf{Q}_t \in \mathbb{R}^{3 \times 3}$: The time-varying process noise covariance matrix, which follows an GARCH(p,q)
- $\boldsymbol{\omega}_0 \in \mathbb{R}^{3 \times 3}$: The baseline (constant) covariance term.
- $\boldsymbol{\alpha}_j \in \mathbb{R}^{3 \times 3}$: The coefficient controlling the influence of past process noise on current volatility.
- $\mathbf{W}_{t-j} \in \mathbb{R}^{3 \times 3}$: The process noise at time $t - j$.
- $\beta_i \in \mathbb{R}^{3 \times 3}$: The persistence coefficient determining how previous covariance $\mathbf{Q}_{t-i}$ affects the current covariance $\mathbf{Q}_t$.

In the following, I use Q_t to represent the state noise covariance in both settings—with or without time-varying volatility. When volatility is not involved, we simply treat $Q_t = Q$ as constant over time.

5 Prediction Step of Kalman Filter

State Prediction

We derived the distribution by calculating the marginal distribution:

$$f(\mathbf{X}_t | \mathbf{Y}_{1:t-1}) = \int \pi(\mathbf{X}_t | \mathbf{X}_{t-1}) \pi(\mathbf{X}_{t-1} | \mathbf{Y}_{1:t-1}) d\mathbf{X}_{t-1}$$

Hence the **state prediction** follows:

$$\mathbf{X}_t | \mathbf{Y}_{1:t-1} \sim \mathcal{MVN}(\hat{\mathbf{X}}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1}) \quad (7)$$

$$\hat{\mathbf{X}}_{t|t-1} = \mathbf{A} \hat{\mathbf{X}}_{t-1|t-1}$$

$$\boldsymbol{\Sigma}_{t|t-1} = \mathbf{A} \boldsymbol{\Sigma}_{t-1|t-1} \mathbf{A}^\top + \mathbf{B} \mathbf{Q}_t \mathbf{B}^\top$$

where

$$\mathbf{Q}_t = \boldsymbol{\omega}_0 + \sum_{j=1}^q \boldsymbol{\alpha}_j \mathbf{W}_{t-j} \mathbf{W}_{t-j}^\top + \sum_{i=1}^p \beta_i \mathbf{Q}_{t-i}$$

Observation Prediction

We derived the distribution by calculating the marginal distribution:

$$f(\mathbf{Y}_t | \mathbf{Y}_{1:t-1}) = \int \pi(\mathbf{Y}_t | \mathbf{X}_t) \pi(\mathbf{X}_t | \mathbf{Y}_{1:t-1}) d\mathbf{X}_t$$

Hence the **observation prediction** follows:

$$\begin{aligned} \mathbf{Y}_t | \mathbf{Y}_{1:t-1} &\sim \mathcal{MVN}(\hat{\mathbf{Y}}_{t|t-1}, \mathbf{F}_t) \\ \hat{\mathbf{Y}}_{t|t-1} &= \mathbf{C}\hat{\mathbf{X}}_{t|t-1} = \mathbf{C}\mathbf{A}\hat{\mathbf{X}}_{t-1|t-1} \\ \mathbf{F}_t &= \mathbf{C}\boldsymbol{\Sigma}_{t|t-1}\mathbf{C}^T + \mathbf{D}\mathbf{R}\mathbf{D}^T = \mathbf{C}(\mathbf{A}\boldsymbol{\Sigma}_{t-1}\mathbf{A}^T + \mathbf{B}\mathbf{Q}_t\mathbf{B}^T)\mathbf{C}^T + \mathbf{D}\mathbf{R}\mathbf{D}^T \end{aligned} \quad (8)$$

State Update

We derived the distribution by calculating the conditional distribution over multivariate normal distribution:

$$f(\mathbf{X}_t | \mathbf{Y}_{1:t}) = \frac{f(\mathbf{Y}_t | \mathbf{X}_t)f(\mathbf{X}_t | \mathbf{Y}_{1:t-1})}{f(\mathbf{Y}_t | \mathbf{Y}_{1:t-1})}$$

Hence the **state update** follows:

$$\begin{aligned} \mathbf{X}_t | \mathbf{Y}_t &\sim \mathcal{MVN}(\hat{\mathbf{X}}_{t|t}, \boldsymbol{\Sigma}_{t|t}) \\ \hat{\mathbf{X}}_{t|t} &= \hat{\mathbf{X}}_{t|t-1} + \mathbf{K}_t(\mathbf{Y}_t - \mathbf{C}\hat{\mathbf{X}}_{t|t-1}) \\ \boldsymbol{\Sigma}_{t|t} &= \boldsymbol{\Sigma}_{t|t-1} - \mathbf{K}_t\mathbf{C}^T\boldsymbol{\Sigma}_{t|t-1} \\ \mathbf{K}_t &= \boldsymbol{\Sigma}_{t|t-1}\mathbf{C}^T\mathbf{F}_t^{-1} = \boldsymbol{\Sigma}_{t|t-1}\mathbf{C}^T(\mathbf{C}\boldsymbol{\Sigma}_{t|t-1}\mathbf{C}^T + \mathbf{D}\mathbf{R}\mathbf{D}^T)^{-1} \end{aligned} \quad (9)$$

6 Parameter Estimation in the Kalman Filter

The Kalman Filter is employed to estimate the latent state vector $\mathbf{X}_t$ in the state-space representation of the Nelson-Siegel model. To fully specify the Kalman Filter, it is necessary to estimate the parameters governing the system dynamics, including the state transition matrix $\mathbf{A}$, and the covariance matrices $\mathbf{Q}_t$ and $\mathbf{R}$. For simplicity, we assume that all three matrices are diagonal.

Meanwhile, the state equation follows an autoregressive (AR) process that includes an intercept term representing the mean, denoted by $\boldsymbol{\mu}$, which is a 3×1 vector. This term should also be incorporated into the specified case of state equation for more accurate estimation.

Since the noise structures have the same dimensions as their respective state (i.e., $\dim(\mathbf{Q}_t) = \dim(\mathbf{X}_t)$ and $\dim(\mathbf{R}) = \dim(\mathbf{Y}_t)$), so we also assume that $\mathbf{B}_t$ and $\mathbf{D}_t$ are identity, which means we can ignore them by the dynamic linear

system.

To capture time-varying volatility in the evolution of the latent factors, we assume that the state error follows a multivariate GARCH(1,1) process. Under these assumptions, we obtain a simplified yet flexible version of the state-space equations suitable for modeling financial time series.

$$\mathbf{X}_t = \mathbf{A} \mathbf{X}_{t-1} + \boldsymbol{\mu} + \mathbf{W}_t, \quad \mathbf{W}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{Q}_t), \quad (10)$$

$$\mathbf{Y}_t = \mathbf{C} \mathbf{X}_t + \mathbf{Z}_t, \quad \mathbf{Z}_t \sim \mathcal{MVN}(\mathbf{0}, \mathbf{R}), \quad (11)$$

$$\mathbf{Q}_t = \boldsymbol{\omega}_0 + \alpha_1 \mathbf{W}_{t-1} \mathbf{W}_{t-1}^\top + \beta_1 \mathbf{Q}_{t-1} \quad (12)$$

6.1 Latent State $\mathbf{X}_t$ Estimation

For each day t , we estimate the latent state vector $\mathbf{X}_t$ using Ordinary Least Squares (OLS) independently. This approach treats the yield curve data at each time point as a cross-sectional regression problem, without assuming a time-dynamic filtering structure during this step. The sequence of estimated latent vectors $\{\hat{\mathbf{X}}_t\}_{t=1}^T$ is then used to fit a multivariate autoregressive (AR) model to capture their temporal dependence.

Using the Ordinary Least Squares, the estimator $\hat{\mathbf{X}}_t$ at each time t is given by:

$$\hat{\mathbf{X}}_t = (\mathbf{C}(\lambda; \boldsymbol{\tau})^\top \mathbf{C}(\lambda; \boldsymbol{\tau}))^{-1} \mathbf{C}(\lambda; \boldsymbol{\tau})^\top \mathbf{Y}_t(\boldsymbol{\tau}) \quad (13)$$

We can expand the linear equation in the matrix form:

$$\begin{bmatrix} y_t(\tau_1) \\ y_t(\tau_2) \\ \vdots \\ y_t(\tau_n) \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} & \frac{1-e^{-\lambda\tau_1}}{\lambda\tau_1} - e^{-\lambda\tau_1} \\ 1 & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} & \frac{1-e^{-\lambda\tau_2}}{\lambda\tau_2} - e^{-\lambda\tau_2} \\ \vdots & \vdots & \vdots \\ 1 & \frac{1-e^{-\lambda\tau_n}}{\lambda\tau_n} & \frac{1-e^{-\lambda\tau_n}}{\lambda\tau_n} - e^{-\lambda\tau_n} \end{bmatrix}}_{\mathbf{C}(\lambda, \boldsymbol{\tau})} \begin{bmatrix} X_t^{(L)} \\ X_t^{(S)} \\ X_t^{(C)} \end{bmatrix} + \begin{bmatrix} W_1^{(L)} \\ W_2^{(L)} \\ W_2^{(L)} \end{bmatrix}.$$

The corresponding residual vector at each time t is:

$$\hat{\mathbf{Z}}_t = \mathbf{Y}_t(\boldsymbol{\tau}) - \mathbf{C}(\lambda; \boldsymbol{\tau}) \hat{\mathbf{X}}_t \quad (14)$$

6.2 State Transition Matrix $\mathbf{A}$ Estimation

The state transition matrix $\mathbf{A}$ is modeled as a **constant diagonal matrix**, capturing the persistence of the Nelson-Siegel factors. This assumption is reasonable, as we can consider L_t , S_t , and C_t to be independent, implying that the

terms can be treated as diagonal and time-invariant.

Since we have the estimated state vectors $\mathbf{X}_t$ from Equation 13, which form a time series of estimated values, we fit each component individually using an autoregressive (AR) process to capture their temporal dependence. The state transition matrix $\mathbf{A}$ is then modeled as a **constant diagonal matrix**, with each diagonal element representing the estimated persistence of the corresponding factor.

We can now expand the state equation in Equation 10 as follows:

$$\begin{bmatrix} \hat{X}_t^{(L)} \\ \hat{X}_t^{(S)} \\ \hat{X}_t^{(C)} \end{bmatrix} = \underbrace{\begin{bmatrix} a^{(L)} & 0 & 0 \\ 0 & a^{(S)} & 0 \\ 0 & 0 & a^{(C)} \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} \hat{X}_{t-1}^{(L)} \\ \hat{X}_{t-1}^{(S)} \\ \hat{X}_{t-1}^{(C)} \end{bmatrix} + \begin{bmatrix} \mu^{(L)} \\ \mu^{(S)} \\ \mu^{(C)} \end{bmatrix} + \begin{bmatrix} W_t^{(L)} \\ W_t^{(S)} \\ W_t^{(C)} \end{bmatrix}$$

Each diagonal element of $\mathbf{A}$ follows an **AR(1) process**:

$$X_t^{(i)} = a^{(i)} X_{t-1}^{(i)} + \mu^{(i)} + W_t^{(i)}, \quad i \in \{L, S, C\} \quad (15)$$

where:

- a_i is the autoregressive coefficient, estimated using the Yule-Walker[Yule, 1927] method.
- μ_i is a drift term capturing the long-run mean dynamics of the state transition process.

To estimate the AR(1) parameters using the Yule-Walker method, we proceed as follows. First, we compute the sample mean $\bar{X}^{(i)}$ of the time series:

$$\bar{X}^{(i)} = \frac{1}{T} \sum_{t=1}^T \hat{X}_t^{(i)}$$

We then subtract the sample mean from the series to obtain the mean-adjusted values: $\tilde{X}_t^{(i)} = X_t^{(i)} - \bar{X}^{(i)}$, and compute the lag-0 and lag-1 autocovariances:

$$\gamma^{(i)}(0) = \frac{1}{T} \sum_{t=1}^T \left(\tilde{X}_t^{(i)} \right)^2, \quad \gamma^{(i)}(1) = \frac{1}{T} \sum_{t=2}^T \tilde{X}_t^{(i)} \tilde{X}_{t-1}^{(i)}$$

The autoregressive coefficient is estimated by:

$$a^{(i)} = \frac{\gamma^{(i)}(1)}{\gamma^{(i)}(0)}$$

Finally, the drift term $\mu^{(i)}$ is recovered using:

$$\mu^{(i)} = \bar{X}^{(i)}(1 - a^{(i)})$$

The corresponding residual at each time t is given by:

$$\hat{W}_t^{(i)} = \hat{X}_t^{(i)} - a^{(i)} \hat{X}_{t-1}^{(i)} - \mu^{(i)} \quad (16)$$

6.3 Observation Noise Covariance Matrix $\mathbf{R}$ Estimation

The observation noise covariance matrix $\mathbf{R}$ is assumed to be constant over time, so we can estimate through the observation residuals 14. Since we have 3 predictors, the degree of freedom is $T-2$:

$$\mathbf{R} = \frac{1}{T-2} \sum_{i=1}^T \mathbf{z}_t \mathbf{z}_t^\top \quad (17)$$

6.4 State Noise Covariance Matrix $\mathbf{Q}$ Estimation (Static)

We input the residual data from Equation 16 at each time t to fit the GARCH process parameters using the **rugarch**

$$\mathbf{Q} = \frac{1}{T-2} \sum_{i=1}^T \mathbf{w}_t \mathbf{w}_t^\top \quad (18)$$

6.5 State Noise Covariance Matrix $\mathbf{Q}_t$ Estimation (Dynamic)

The state noise covariance matrix $\mathbf{Q}_t$ governs the stochastic evolution of the latent state variables. To allow for time-varying volatility, the state noise structure is modeled using a multivariate GARCH(1,1) process. For simplicity, we assume that $\mathbf{Q}_t$ is a diagonal matrix, reflecting the assumption that there is no contemporaneous correlation between the latent parameters L_t , S_t , and C_t . From the equation 12, can model $\mathbf{Q}_t$ as follow:

$$\mathbf{Q}_t = \begin{bmatrix} q_t^{(L)} & 0 & 0 \\ 0 & q_t^{(S)} & 0 \\ 0 & 0 & q_t^{(C)} \end{bmatrix}$$

Each diagonal element follows a **GARCH(1,1) process**:

$$q_t^{(i)} = \omega_0^{(i)} + \alpha_1^{(i)} (w_{t-1}^{(i)})^2 + \beta_1^{(i)} q_{t-1}^{(i)}, \quad i \in \{L, S, C\} \quad (19)$$

From all parameter fitted above, we have the fitted Kalman filter paramter, and we can fit all the Latent state like equation 7. Thus, we have the predicted state $\mathbf{X}_t | \mathbf{Y}_{t-1}$ from equation 7 and the filtered state $\mathbf{X}_t | \mathbf{Y}_t$ from equation 9. Then we consider the state residual:

$$\mathbf{Z}_t = \mathbf{X}_t | \mathbf{Y}_{t-1} - \mathbf{X}_t | \mathbf{Y}_t \quad (20)$$

We input the residual data from Equation 20 at each time t to fit the GARCH process parameters using the **rugarch** package[Ghalanos, 2015]. The package uses recursive maximum likelihood estimation to estimate the parameters, returning the optimal estimates. These estimated parameters can then be utilized within the Kalman Filter framework under GARCH(1,1) setting.

This matrix is estimated by computing the sample covariance of residuals obtained after an initial Kalman Filter run with preliminary parameter estimates.

7 Case Studies

In this case study, we use the same dataset from the secondary markets, where the St. Louis Federal Reserve [of the Treasury, 2021] collects par yield curve data. This dataset is utilized to run a Kalman Filter, applying the previously described parameter estimation procedures to make yield rate predictions.

In this study, we fit two models. The first is the traditional Kalman Filter (Equation 4, 5), which assumes constant volatility over time, and we will call this model **FK** for convinience. The second model allows for time-varying volatility in the state equation noise, implemented by incorporating a GARCH specification, resulting in a Kalman Filter with GARCH-based state noise (Equation 10, 11, 12). We will call the second model **KF-GARCH**.

Table 1 presents the predicted yield curves from two models: **KF** and **KF-GARCH**. The predictions span five business days, from 2024-11-18 to 2024-11-22, and include yields at five different maturities .

At a glance, both models produce similar forecasts, particularly at shorter maturities. However, the KF-GARCH model introduces subtle differences—especially at longer maturities—where it slightly adjusts predictions to reflect changes in estimated volatility. This improved responsiveness is due to the incorporation of GARCH-driven state noise, which captures time-varying uncertainty in the latent factors.

To evaluate model performance, we compare the log-likelihood values:

- **Kalman Filter:** Log-Likelihood = -1081.513
- **Kalman Filter-GARCH:** Log-Likelihood = -1033.116

The KF-GARCH model yields a higher (less negative) log-likelihood, indicating a better fit to the observed yield data. This improvement suggests that accounting for volatility clustering in the state dynamics enhances the model’s ability to capture the behavior of the yield curve.

Figure 1 compares the yield curve predictions from the standard Kalman Filter (KF) and the volatility-augmented Kalman Filter (KF-GARCH) across four maturities: 5, 10, 20, and 30 years. Across all maturities, both models are able

Model	Date	1 Month	5 Years	10 Years	20 Years	30 Years
KF	2024-11-18	4.505397	4.384999	4.427179	4.546191	4.693423
	2024-11-19	4.501594	4.381933	4.424346	4.543321	4.690388
	2024-11-20	4.486415	4.362522	4.402861	4.521018	4.667940
	2024-11-21	4.489876	4.365829	4.405740	4.523370	4.669759
	2024-11-22	4.502518	4.379885	4.420042	4.537317	4.683087
KF-GARCH	2024-11-18	4.543090	4.376839	4.417955	4.558202	4.735888
	2024-11-19	4.550408	4.374264	4.412120	4.552683	4.732406
	2024-11-20	4.542234	4.350370	4.381355	4.520047	4.700479
	2024-11-21	4.547410	4.361667	4.393436	4.530197	4.707523
	2024-11-22	4.560762	4.380397	4.412915	4.548063	4.722752
Real Value	2024-11-18	4.584090	4.291318	4.424253	4.579328	4.699603
	2024-11-19	4.575160	4.263728	4.393899	4.545012	4.659659
	2024-11-20	4.575935	4.285405	4.415384	4.557056	4.659406
	2024-11-21	4.590393	4.309536	4.434127	4.575709	4.679592
	2024-11-22	4.592643	4.309326	4.415259	4.557571	4.669450

Table 1: Yield Rate Predictions: KF vs. KF-GARCH vs. Real

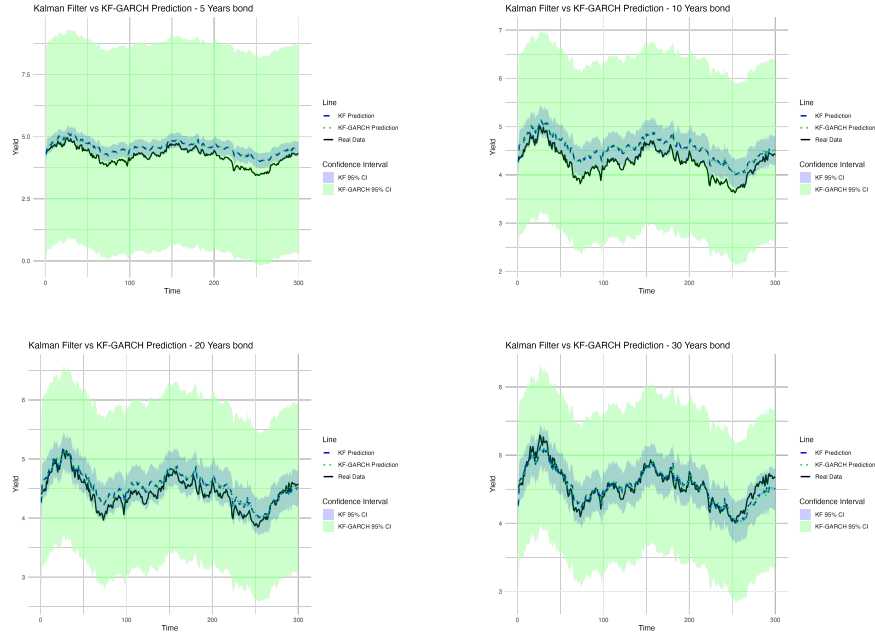


Figure 1: Yield Rate Prediction using KF and KF-GARCH

to track the general trend of the observed yields. However, notable differences emerge in how each model represents uncertainty:

- The standard KF model assumes constant volatility, resulting in confidence bands of fixed width. These may understate uncertainty during periods of market turbulence.
- The KF-GARCH model, by contrast, incorporates time-varying volatility into the state equation. This leads to wider and more adaptive confidence intervals that respond dynamically to changes in the level of market uncertainty and account for higher-order characteristics such as kurtosis.

For the 5-year and 10-year bond, the Kalman Filter and KF-GARCH models produce similar point predictions; however, the KF-GARCH model offers more conservative confidence bands that expand during yield dips, reflecting increased uncertainty that is not captured by standard Kalman Filter. In the 20-year bond, both models track the real yield closely, but KF-GARCH exhibits wider confidence intervals during upward movements, demonstrating its sensitivity to changes in volatility. Finally, in the 30-year bond, KF-GARCH more effectively captures fluctuations in yield rate, and the divergence in confidence intervals becomes more pronounced. This highlights the growing importance of modeling time-varying volatility at longer maturities, where market dynamics tend to be more unstable.

In conclusion, while both models are effective in forecasting yields, the KF-GARCH model offers a more realistic and flexible representation of prediction uncertainty. Its ability to model volatility clustering makes it particularly well-suited for financial applications involving dynamic risk environments.

8 Conclusion and Future Studies

In this study, we explored the use of the Kalman Filter for modeling and predicting the U.S. Treasury yield curve through the Dynamic Nelson Siegel framework. By comparing a traditional Kalman Filter (KF) model with a volatility-adjusted version using GARCH dynamics (KF-GARCH), we assessed the importance of incorporating time-varying volatility into the state space model.

The results demonstrate that both models effectively capture the general trend of the yield curve across multiple maturities. However, the KF-GARCH model provides more adaptive and realistic confidence intervals, particularly under volatile market conditions. The integration of a multivariate GARCH(1,1) process into the state noise structure allows the model to dynamically adjust its uncertainty estimates. This flexibility is especially valuable in financial markets, where regime changes and volatility clustering are common.

Future Studies

While the current implementation demonstrates strong empirical performance, several extensions can be considered in future research:

- **Modeling Cross-Correlations:** Future work can relax the diagonal assumptions on the covariance matrices to account for potential correlations between the level, slope, and curvature factors.
- **Parameter Estimation:** Employing more advanced parameter estimation methods, such as the Expectation-Maximization algorithm, could improve the stability and accuracy of the estimated model parameters.
- **Bayesian Estimation:** Bayesian Kalman filtering techniques or particle filtering methods could be employed to better handle parameter uncertainty and non-Gaussian errors.

Overall, the Kalman Filter combined with time-varying volatility modeling offers a powerful framework for tracking and forecasting term structures in financial markets, laying the foundation for more advanced dynamic risk modeling in future work.

References

- [Christensen et al., 2011] Christensen, J. H., Diebold, F. X., and Rudebusch, G. D. (2011). The affine arbitrage-free class of nelson–siegel term structure models. *Journal of Econometrics*, 164(1):4–20.
- [Diebold and Li, 2006] Diebold, F. X. and Li, C. (2006). Forecasting the term structure of government bond yields. *Journal of econometrics*, 130(2):337–364.
- [Fama and Bliss, 1987] Fama, E. F. and Bliss, R. R. (1987). The information in long-maturity forward rates. *The American Economic Review*, pages 680–692.
- [Ghalanos, 2015] Ghalanos, A. (2015). *rugarch: Univariate GARCH models*. R package version 1.4-6. Available from: <https://CRAN.R-project.org/package=rugarch>.
- [McCulloch, 1971] McCulloch, J. H. (1971). Measuring the term structure of interest rates. *the Journal of Business*, 44(1):19–31.
- [Nelson and Siegel, 1987] Nelson, C. R. and Siegel, A. F. (1987). Parsimonious modeling of yield curves. *Journal of business*, pages 473–489.
- [of the Treasury, 2021] of the Treasury, U. D. (2021). Yield curve methodology change information sheet. Technical report, U.S. Department of the Treasury. Available at: <https://home.treasury.gov/policy-issues/financing-the-government/yield-curve-methodology-change-information-sheet>.
- [Shea, 1984] Shea, G. S. (1984). Pitfalls in smoothing interest rate term structure data: Equilibrium models and spline approximations. *Journal of Financial and Quantitative Analysis*, 19(3):253–269.
- [Steeley, 1991] Steeley, J. M. (1991). Estimating the gilt-edged term structure: Basis splines and confidence intervals. *Journal of Business Finance & Accounting*, 18(4).
- [Yule, 1927] Yule, G. U. (1927). On a method of investigating periodicities in disturbed series, with special reference to the sunspot numbers. *Philosophical Magazine*, 2(11):78–87.